

• Hugging Face • Stanford University

SmolVLM: Redefining Small and Efficient Multimodal Models

Compact vision-language models achieving competitive performance at sub-1GB RAM through strategic architectural optimizations

arXiv 2504.05299

3 Model Sizes: 256M • 500M • 2.2B

Fully Open-Source

02 Large VLMs Are Powerful but Impractical for Edge

STATE-OF-THE-ART VLM

27.7 GB

Molmo-A1B-7B GPU RAM at batch=1

TYPICAL MOBILE BUDGET

< 4 GB

Total device RAM for all processes

PARAMETER GAP

80B+

Idefics-80B, the model SmolVLM-256M surpasses

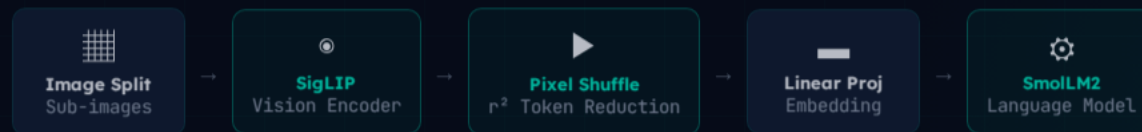
SMOLVLM-256M

0.8 GB

Runs on smartphones · 300× fewer params than Idefics-80B

The challenge: Deploy multimodal intelligence on edge devices where GPU memory is scarce and latency budgets are tight.

03 Architecture: Image Splitting → Pixel Shuffle → SmolLM2



Finding 1

Balanced Encoder-LM Compute

Smaller SigLIP-B/16 (93M) outperforms SigLIP-SO400M (428M) for compact LMs

Finding 2

Extended Context Length

2K – 16K tokens via RoPE base scaling enables higher-resolution inputs

Finding 3

Aggressive Token Compression

Pixel shuffle $r=4$ yields 16× token reduction for small models

04

Smaller Vision Encoders Outperform Larger Ones

Avg. benchmark score by encoder-LM pairing (SmolVLM family)

SIGLIP-B/16 · 93M PARAMS



SIGLIP-S0400M · 428M PARAMS



05 Pixel Shuffle (r=4) Cuts Tokens 16x

Before: 4x4 spatial = 16 tokens



↓ space-to-depth (r=4)



After: 1x1 spatial = 1 token (16 channels)

16x token reduction

Mechanism

Space-to-depth rearrangement

Formula

Tokens reduced by r^2 ($r=4 \rightarrow 16\times$)

Small Models (256M, 500M)

r=4 preferred — eases attention overhead

06 **Context Extension: 2K → 16K Tokens**

Context window scaling by model variant



RoPE base adjustment



Technique
RoPE base 10K → 273K

Fine-tuning
Mixed long + short-context data

256M & 500M
8K context length

07

Competitive Performance at Fractional RAM

Category	Benchmark	SmolVLM 256M	SmolVLM 500M	SmolVLM 2.2B	Best Efficient OS
SINGLE-IMAGE UNDERSTANDING					
	OCRBench	52.6%	61.0%	72.9%	54.7% (HoLmo)
	AI2D	46.4%	59.2%	70.0%	71.0% (HoLmo)
	ChartQA	55.6%	62.8%	68.7%	48.0% (HoLmo)
	TextVQA	50.2%	60.2%	73.0%	61.5% (HoLmo)
	DocVQA	58.3%	70.5%	80.0%	77.7% (HoLmo)
	ScienceQA	73.8%	80.0%	89.6%	87.5% (HoLmo)
MULTI-TASK REASONING					
	MMU	29.0%	33.7%	42.0%	33.9% (HoLmo)
	MathVista	35.9%	40.1%	51.5%	37.6% (HoLmo)
	MMStar	34.6%	38.3%	46.0%	43.1% (HoLmo)
VIDEO COMPREHENSION					
	Video-MME	33.7%	42.2%	52.1%	45.0% (InternVL2)
	MLVU	40.6%	47.3%	55.2%	48.2% (InternVL2)
Average (All)		44.0%	51.0%	59.8%	-

SmolVLM-256M 0.8 GB

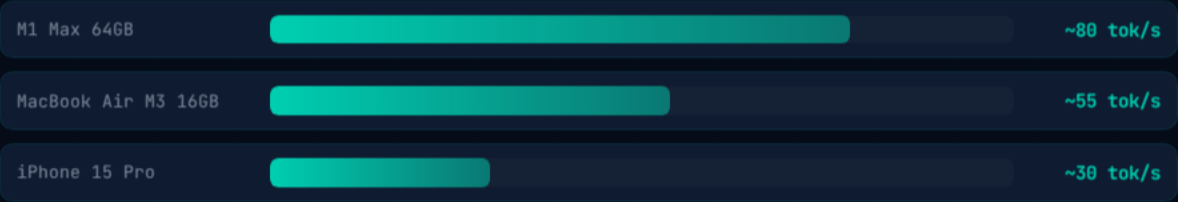
SmolVLM-500M 1.2 GB

SmolVLM-2.2B 4.9 GB

HoLmo-A1B-7B 27.7 GB

08 **On-Device: Up to 80 Tokens/sec on Apple Silicon**

SmolVLM-256M throughput (batch=1)



High-batch throughput (batch=64)



Deployment Platform

HuggingSnap iOS App

Photos & video processed entirely on-device, zero cloud dependency

Peak Throughput

~1000 tok/s

SmolVLM-256M at batch=64 on M1 Max 64GB

09 Design Principles for Efficient Multimodal Models

01

Balance Encoder-LM Compute

Smaller vision encoders (93M) outperform larger ones (428M) for compact LMs — allocate parameters where they matter most.

02

Extend Context Length Aggressively

Scaling from 2K to 8K-16K via RoPE adjustment unlocks higher image resolution and richer multimodal understanding.

03

Use Aggressive Token Compression

Pixel shuffle $r=4$ (16× reduction) is optimal for small models — eases attention overhead and improves long-context modeling.

04

Sub-1GB Inference Is Achievable

SmolVLM-256M runs at 0.8 GB RAM and surpasses the 300× larger LLaVA-72B, proving architecture > raw scale.

Efficient multimodal models are not a compromise — they are a **design discipline**.